

Chapter 14: Solutions



Dr. Wang

Learning Objectives

- Determine the ΔH when solutions are mixed based upon a temperature change that occurs when two liquids are mixed.
- Convert between different units of concentration.
- Determine the van't Hoff factor (i) for a solute based on the solute.
- Calculate the difference between a solution and the pure substance (solvent) on vapor pressure, boiling point, or freezing point.
- Describe why the vapor pressure of a solution is lower than the vapor pressure of the pure substance.
- Describe why the boiling point of a solution is higher than the boiling point of the pure substance.
- Describe why the freezing point of a solution is lower than the freezing point of the pure substance.

Solution

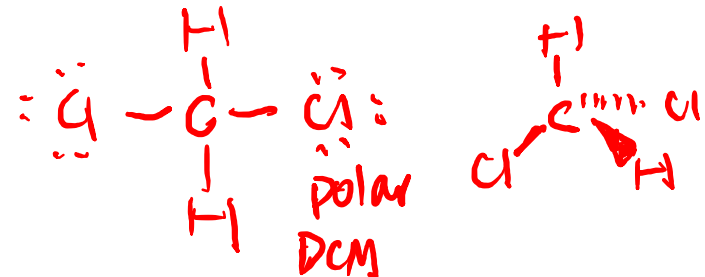
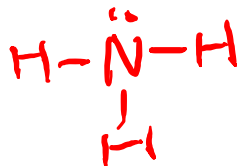
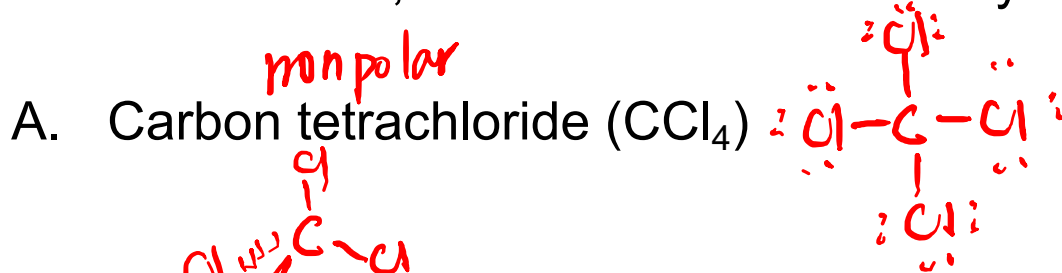
- **Solution** is a homogenous mixture with 2 or more components
- **Solution** can be a (s), a (l) or a (g)
- **Solute** (the minor component); **Solvent** (the major component)
- Aqueous solution: **Solute** ((s), (l), (g)); **Solvent** (H₂O(l))
- **Miscibility**: "Like dissolves Like"

Polar solutes can be dissolved in polar solvent.

Nonpolar solutes can be dissolved in non polar solvent.

→ ionic

In which solvent, KCl has a better solubility?



Solution

- **Solution** is a homogeneous mixture with 2 or more components
- **Solution** can be a solid, a liquid or a gas
- **Solute** (the minor component); **Solvent** (the major component)
aqueous solution: solutes (solid, liquid, gas) solvent (H₂O)
- **Miscibility**: “Like dissolves Like”
 - Polar solutes can be dissolved in polar solvent.
 - Nonpolar solutes can be dissolved in non polar solvent.

In which solvent, KCl has a better solubility?

A. Carbon tetrachloride (CCl₄)

B Ammonia (NH₃)



Concentration

- **Concentration** ($\frac{\text{solute}}{\text{solution / solvent}}$) is the ratio between mol of a **solute** within Volume or mass or _____ of **solution** or **solvent**.
- Molar**ity** $M = \frac{n_{\text{solute}} \text{ (in mol)}}{V_{\text{solution}} \text{ (in L)}} \text{ (unit: } \frac{\text{mol}}{\text{L}}, M \text{)}$
- Molal**ity** $m = \frac{n_{\text{solute}} \text{ (in mol)}}{\text{mass}_{\text{solvent}} \text{ (in kg)}} \text{ (unit: } \frac{\text{mol}}{\text{kg}}, m \text{)}$

Learning Check

Assume 1L of solution

Convert 2.45 M aqueous solution of methanol (CH_3OH ; $m_m = 32.04 \text{ g/mol}$) with the density of 0.976 g/mL to molality. ($m_{\text{H}_2\text{O}} = 18.02 \text{ g/mol}$)

solution

$$m_{\text{CH}_3\text{OH}} = \frac{2.45 \text{ mol}}{898 \text{ g} \times \frac{1 \text{ kg}}{1000 \text{ g}}} = \boxed{2.73 \text{ m}} \quad \boxed{2.45 \text{ M}}$$

$$\text{mol}(\text{CH}_3\text{OH}) = \frac{2.45 \text{ mol CH}_3\text{OH}}{1 \text{ L}} \times 1 \text{ L} = 2.45 \text{ mol CH}_3\text{OH}$$

$$m(\text{H}_2\text{O}) = \frac{0.976 \frac{\text{g}}{\text{mL}} \times 1 \text{ L} \times \frac{1 \text{ m}^3}{10^{-3} \text{ L}}}{m(\text{CH}_3\text{OH})} - 2.45 \text{ mol} \times \frac{32.04 \text{ g}}{1 \text{ mol}} = 898 \text{ g H}_2\text{O}$$

Concentration

- **Concentration** ($\frac{\text{solute (mol, V, mass...)}}{\text{solvent (mol, V, mass...)}}$) is the ratio between mol of a **solute** within mol or volume or mass of **solution** or **solvent**.
- Molar**ity** $M = \frac{n_{\text{solute (in mol)}}}{V_{\text{solution (in L)}}$ (unit: mol/L) M
- Molal**ity** $m = \frac{n_{\text{solute (in mol)}}}{\text{mass}_{\text{solvent (in kg)}}$ (unit: mol/kg) m

Learning Check

Convert 2.45 M aqueous solution of methanol (CH_3OH ; $m_m = 32.04 \text{ g/mol}$) with the density of 0.976 g/mL to molality. ($m_{\text{H}_2\text{O}} = 18.02 \text{ g/mol}$)

$$m_{\text{CH}_3\text{OH}} = \frac{2.45 \text{ mol CH}_3\text{OH}}{1 \text{ L solution}} \times \frac{1 \text{ mL}}{0.976 \text{ g}} \times \frac{10^{-3} \text{ L}}{1 \text{ mL}} \times \frac{10^3 \text{ g}}{1 \text{ kg}} = \frac{2.51 \text{ mol CH}_3\text{OH}}{1 \text{ kg solution}}$$

$$= 2.51 \text{ m}$$

Concentration

- **Concentration** ($\frac{\text{solute}}{\text{solution or solvent}}$) is the ratio between mol number of a **solute** within volume or mass or **mol number** of **solution** or **solvent**.
- Molarity $M = \frac{n_{\text{solute}} (\text{in } \underline{\quad})}{V_{\text{solution}} (\text{in } \underline{\quad})}$ (unit: $\underline{\quad}$)
- Mole Fraction: $X_A = \frac{n_{\text{particle of solute}} (\text{in } \underline{\text{mol}})}{\sum n_{\text{particles in solutions}} (\text{in } \underline{\text{mol}})} = \frac{n_A}{n_A + n_B + \dots}$ (unit: no unit)

Learning Check *Assume 1 L solution*

Convert 2.45 M aqueous solution of methanol (CH_3OH ; $\text{mm} = 32.04 \text{ g/mol}$) with the density of 0.976 g/mL to mole fraction. ($\text{mm}_{\text{H}_2\text{O}} = 18.02 \text{ g/mol}$)

2.45 mol CH_3OH

$$898 \text{ g H}_2\text{O} \times \frac{1 \text{ mol}}{18.02 \text{ g}} = 49.8 \text{ mol H}_2\text{O}$$

$$X_{\text{CH}_3\text{OH}} = \frac{2.45 \text{ mol}}{2.45 \text{ mol} + 49.8 \text{ mol}} = 0.0469 < 0.5$$

Concentration

- **Concentration** ($\frac{\text{solute}}{\text{solution or solvent}}$) is the ratio between mol number of a **solute** within volume or mass or ml number of **solution** or **solvent**.
- Molarity $M = \frac{n_{\text{solute}} \text{ (in mol)}}{V_{\text{solution}} \text{ (in L)}}$ (unit: M)
- Mole Fraction: $X_A = \frac{n_{\text{particle of solute}} \text{ (in mol)}}{\sum n_{\text{particles in solutions}} \text{ (in mol)}} = \frac{n_A}{n_A + n_B + \dots}$ (unit: no unit)

Learning Check

Convert 2.45 *M* aqueous solution of methanol (CH_3OH ; mm = 32.04 g/mol) with the density of 0.976 g/mL to mole fraction. (mm_{H₂O} = 18.02 g/mol)

Assume 1L solution:

$$n_{\text{CH}_3\text{OH}} = \frac{2.45 \text{ mol CH}_3\text{OH}}{1 \text{ L}} \times 1 \text{ L} = 2.45 \text{ mol CH}_3\text{OH}$$

$$m_{\text{solution}} = 0.976 \frac{\text{g}}{\text{mL}} \times 1 \text{ L} \times \frac{1 \text{ mL}}{10^{-3} \text{ L}} = 976 \text{ g}$$

$$n_{\text{H}_2\text{O}} = 898 \text{ g} \times \frac{1 \text{ mol}}{18.02 \text{ g}} = 49.8 \text{ mol}$$

$$m_{\text{H}_2\text{O}} = 976 \text{ g} - 2.45 \text{ mol} \times \frac{32.04 \text{ g}}{1 \text{ mol}} = 898 \text{ g}$$

$$X_{\text{CH}_3\text{OH}} = \frac{2.45 \text{ mol}}{49.8 \text{ mol} + 2.45 \text{ mol}} = 0.0469$$

Your Turn!

A solution contains 25.0 g acetone ($\text{C}_3\text{H}_6\text{O}$) and 75.0g water. Density of solution = 0.985 g/mL, $\text{mm}_{\text{acetone}} = 58.08 \text{ g/mol}$ and $\text{mm}_{\text{H}_2\text{O}} = 18.02 \text{ g/mol}$. Calculate the concentration in:

1. Molarity:

4.24 M

2. Molality:

5.74 m

3. Mole Fraction:

0.0937

Concentration Unit	Abbreviation	Concentration Formula
Molarity	M	$M = \frac{\text{moles of solute}}{\text{Liters of solution}}$
molality	m	$m = \frac{\text{moles of solute}}{\text{kg of solvent}}$
Mole Fraction	X	$X_A = \frac{\text{moles of A}}{\text{total \# moles in solution}} = \frac{n_A}{n_A + n_B + n_C + \dots}$

~~3.00937~~
~~2.574 m~~
~~1.424 M~~

Your Turn!

A solution contains 25.0 g acetone ($\text{C}_3\text{H}_6\text{O}$) and 75.0g water. Density of solution=0.985 g/mL, $\text{mm}_{\text{acetone}} = 58.08 \text{ g/mol}$ and $\text{mm}_{\text{H}_2\text{O}} = 18.02 \text{ g/mol}$. Calculate the concentration in:

1. Molarity:
$$\frac{25.0 \text{ g} \times \frac{1 \text{ mol}}{58.08 \text{ g}}}{(25.0 \text{ g} + 75.0 \text{ g}) \times \frac{1 \text{ mL}}{0.985 \text{ g}} \times \frac{10^{-3} \text{ L}}{1 \text{ mL}}} = 4.24 \text{ M}$$

2. Molality:
$$\frac{25.0 \text{ g} \times \frac{1 \text{ mol}}{58.08 \text{ g}}}{75.0 \text{ g} \times \frac{1 \text{ kg}}{1000 \text{ g}}} = 5.74 \text{ m}$$

3. Mole Fraction:
$$\frac{25.0 \text{ g} \times \frac{1 \text{ mol}}{58.08 \text{ g}}}{25.0 \text{ g} \times \frac{1 \text{ mol}}{58.08 \text{ g}} + 75.0 \text{ g} \times \frac{1 \text{ mol}}{18.02 \text{ g}}} = \frac{0.427}{0.427 + 4.16} = 0.0937$$

Concentration Unit	Abbreviation	Concentration Formula
Molarity	M	$M = \frac{\text{moles of solute}}{\text{Liters of solution}}$
molality	m	$m = \frac{\text{moles of solute}}{\text{kg of solvent}}$
Mole Fraction	X	$X_A = \frac{\text{moles of A}}{\text{total \# moles in solution}} = \frac{n_A}{n_A + n_B + n_C + \dots}$

1. 7.58 M
2. 9.51 m
3. 0.146

Announcements 3/16

Exam 2 at 11 AM on 3/20:

- 50 minutes long
- Calculators (any kind), pens/pencils/highlighters, and erasers are allowed. *rule*
- Bonus point for Unit 2:

2nd Chem RePS or SI worksheet: due 11:59 PM 3/18

3rd Chem RePS or SI worksheet: due 10 AM 3/20

Assignments that are due soon

- Pre-Class Assignment Week 7-W

Office Hour

- Dr. Wang's: Tue 10-11 AM at KC 230
- Paige Kwan's: Mon 11:30 AM -12:30 PM at the TLC (Demille 160P)
- Saige Douglass's: Tue ~~5:30-6:30~~ ^{12:00-1:00} PM at the TLC (Demille 160P)

Zoom exam 2 Review by SI (Amelia & Joana)

5-6:30 PM 3/16

Common Monoatomic Cations and Anions

Charge for common Polyatomic Ions:

$(\text{NH}_4)^{+1}$	$(\text{OH})^{-}$
$(\text{C}_2\text{H}_3\text{O}_2)^{-1}$	$(\text{NO}_3)^{-}$
$(\text{CO}_3)^{2-}$	$(\text{SO}_4)^{2-}$
$(\text{HCO}_3)^{-}$	$(\text{PO}_4)^{3-}$

Main group elements tend to lose/gain e^- to obtain same e^- with the closest noble gas.

Main Group Metal:

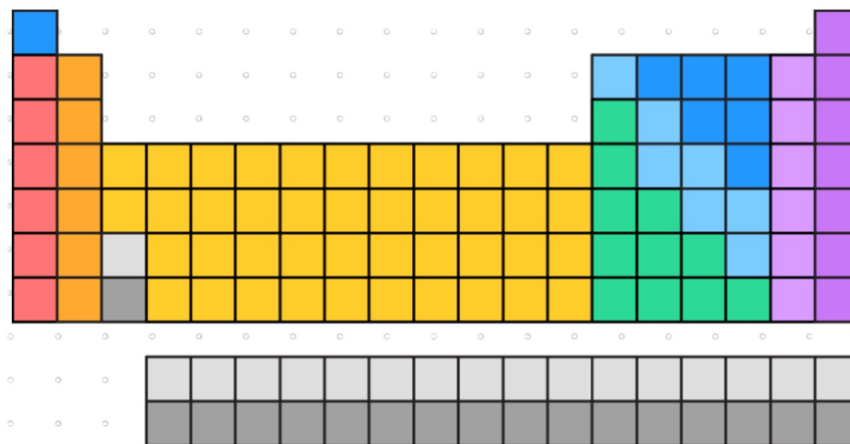
Group 1 forms +1 cations, Group 2 forms +2 cations, Al forms +3 cations

- (Inner/_/Post)Transition metal can form multiple cations with different charge

Nonmetal:

Group 15 forms -3 anions, Group 16 forms -2 anions, Group 17 forms -1 anions

Molality and Mole Fraction of Ionic compounds in water



Dissociation of MgCl_2 :



1 mol of MgCl_2 has 1 mol of Mg^{2+} and 2 mol of Cl^{-} .

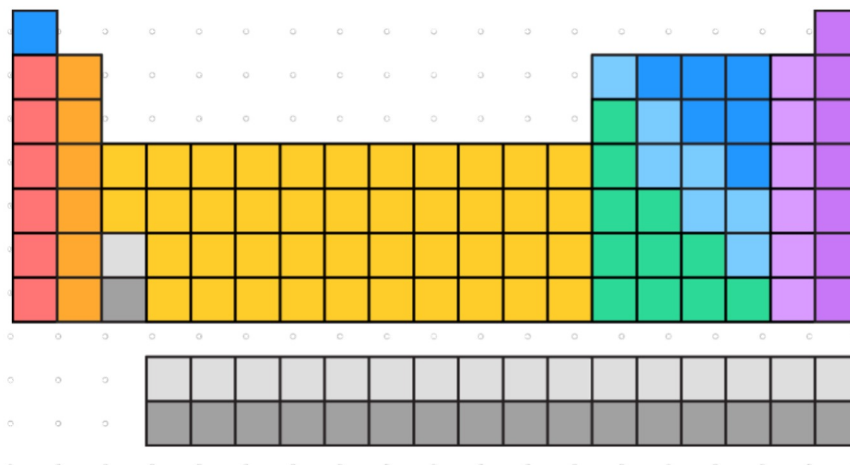
The total particle number (including Mg^{2+} and Cl^{-}) in 1 mol of MgCl_2 is 3 mol

A solution contain 9.52 g of MgCl_2 (0.100 mol of MgCl_2) and 100. g water (5.55 mol of water).

$$m(\text{Cl}^{-}) = \frac{0.200 \text{ mol Cl}^{-}}{100. \text{ g} \times \frac{1 \text{ kg}}{10^3 \text{ g}}} = 2.00 \frac{\text{mol}}{\text{kg}} = 2.00 \text{ m}$$

$$X(\text{Cl}^{-}) = \frac{0.200 \text{ mol Cl}^{-}}{5.5 \text{ mol H}_2\text{O} + 0.200 \text{ mol Cl}^{-} + 0.100 \text{ mol Mg}^{2+}} = 0.0342$$

Molality and Mole Fraction of Ionic compounds in water



Dissociation of MgCl_2 :



1 mol of MgCl_2 has 1 mol of Mg^{2+} and 2 mol of Cl^- .

The total particle number (including Mg^{2+} and Cl^-) in 1 mol of MgCl_2 is 3 mol

A solution contain 9.52 g of MgCl_2 (0.1 mol of MgCl_2) and 100.00 g water (5.55 mol of water).

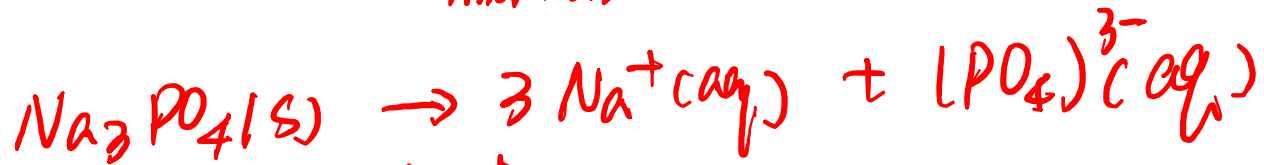
$$m(\text{Cl}^-) = \frac{0.1 \text{ mol MgCl}_2 \times \frac{2 \text{ Cl}^-}{1 \text{ MgCl}_2}}{100.00 \text{ g} \times \frac{1 \text{ kg}}{10^3 \text{ g}}} = 2 \text{ m}$$

$$X(\text{Cl}^-) = \frac{0.1 \text{ mol MgCl}_2 \times \frac{2 \text{ Cl}^-}{1 \text{ MgCl}_2}}{0.1 \text{ mol MgCl}_2 \times \frac{1 \text{ Mg}^{2+} + 2 \text{ Cl}^-}{1 \text{ MgCl}_2} + 5.55 \text{ mol H}_2\text{O}} = \frac{0.200}{0.300 + 5.55} = 0.0342$$

Your Turn!

A solution contains 5.00 g Na_3PO_4 (0.0305 mol of Na_3PO_4) and 100. g water (5.55 mol of water). Calculate the concentration of Na^+ in Molality and mol fraction in this solution.

$$0.0305 \text{ mol } \text{Na}_3\text{PO}_4 \times \frac{3 \text{ mol } \text{Na}^+}{1 \text{ mol } \text{Na}_3\text{PO}_4} = 0.0915 \text{ mol } \text{Na}^+$$



$$m(\text{Na}^+) = \frac{0.0915 \text{ mol } \text{Na}^+}{0.1 \text{ kg H}_2\text{O}} = 0.915 \text{ m}$$

$$X(\text{Na}^+) = \frac{0.0915 \text{ mol } \text{Na}^+}{5.55 \text{ mol H}_2\text{O} + 0.0915 \text{ mol } \text{Na}^+ + 0.0305 \text{ mol } \text{PO}_4^{3-}} = 0.0161$$

Concentration Unit	Abbreviation	Concentration Formula
Molarity	M	$M = \frac{\text{moles of solute}}{\text{Liters of solution}}$
molality	m	$m = \frac{\text{moles of solute}}{\text{kg of solvent}}$
Mole Fraction	X	$X_A = \frac{\text{moles of A}}{\text{total \# moles in solution}} = \frac{n_A}{n_A + n_B + n_C + \dots}$

Your Turn!

A solution contains 5.00 g Na_3PO_4 (0.0305 mol of Na_3PO_4) and 100. g water (5.55 mol of water). Calculate the concentration of Na^+ in Molality and mol fraction in in this solution.

$$m = \frac{0.0305 \text{ mol } \text{Na}_3\text{PO}_4 \times \frac{3 \text{ Na}^+}{1 \text{ Na}_3\text{PO}_4}}{100.0 \text{ g} \times \frac{1 \text{ kg}}{10^3 \text{ g}}} = 0.915 \text{ m}$$

$$X = \frac{0.0305 \text{ mol } \text{Na}_3\text{PO}_4 \times \frac{3 \text{ Na}^+}{1 \text{ Na}_3\text{PO}_4}}{0.0305 \text{ mol } \text{Na}_3\text{PO}_4 \times \frac{3 \text{ Na}^+ + 1 \text{ PO}_4^{3-}}{1 \text{ Na}_3\text{PO}_4} + 5.55 \text{ mol H}_2\text{O}} = 0.0161$$

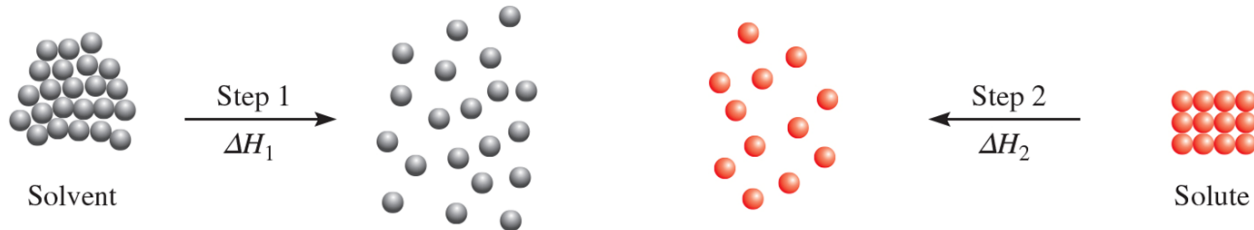
Concentration Unit	Abbreviation	Concentration Formula
Molarity	M	$M = \frac{\text{moles of solute}}{\text{Liters of solution}}$
molality	m	$m = \frac{\text{moles of solute}}{\text{kg of solvent}}$
Mole Fraction	X	$X_A = \frac{\text{moles of A}}{\text{total \# moles in solution}} = \frac{n_A}{n_A + n_B + n_C + \dots}$

0.915 m, 0.0161

Why do Solutions Form?

$$\Delta G = \Delta H - T\Delta S$$

- A spontaneous process favors (+) of entropy ΔS and (-) of enthalpy ΔH .
- When a solution forms: ΔS (+) and ΔH ?



Lattice energy
ionic bond

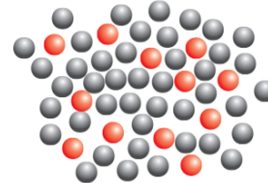
Step 1: Separating the solvent particles

break IMFs
 $\Delta H_{\text{solvent}}$ (+)

Step 3: ΔH_3

Step 2: Separating the solute particles

break IMFs
 ΔH_{solute} (+)



Solution

Step 3: Mix the solute's and solvent's particles

form IMFs
 ΔH_{mix} (-)

$$\Delta H_{\text{solution}} = \Delta H_{\text{solvent}} + \Delta H_{\text{solute}} + \Delta H_{\text{mix}} = \Delta H_{\text{hydration}} + \Delta H_{\text{solute}}$$

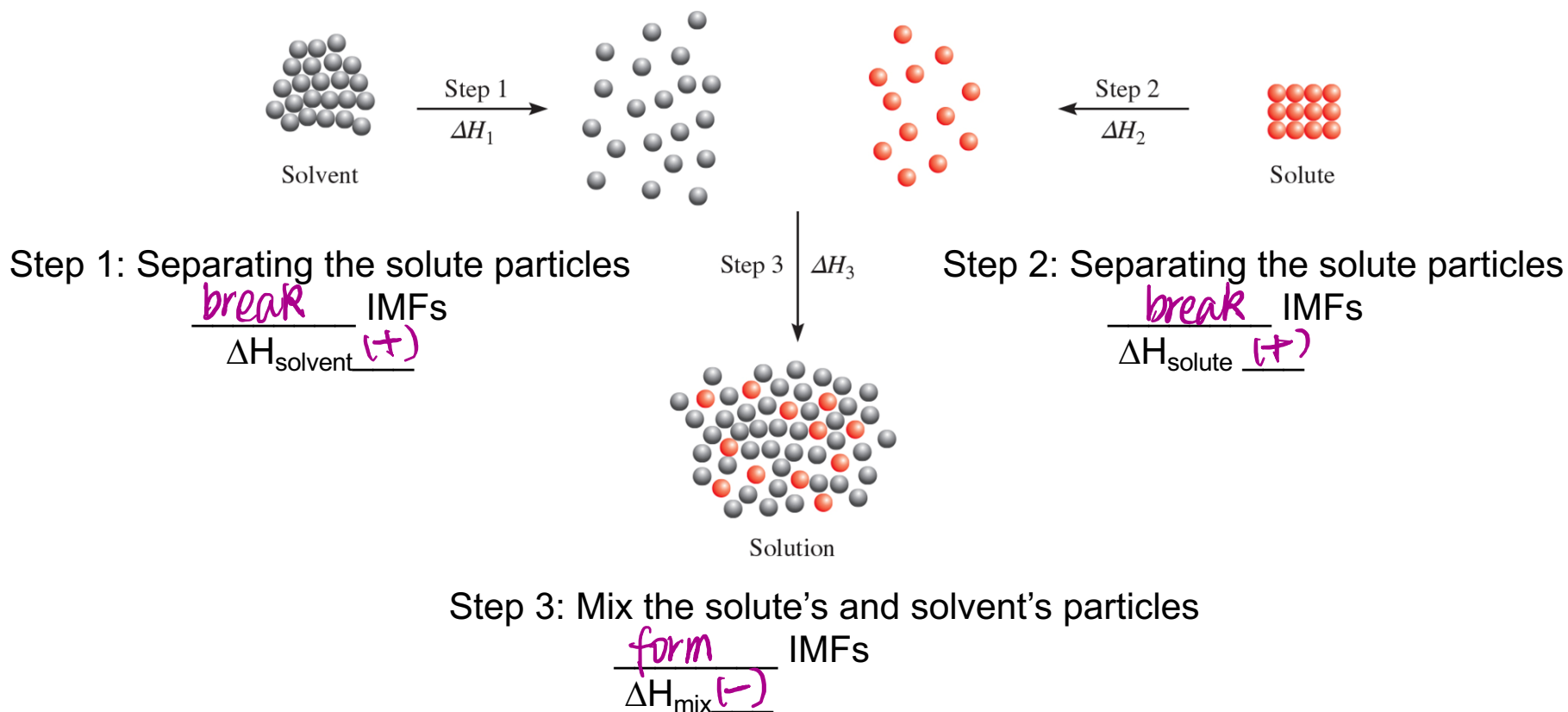
$$\Delta H_1 + \Delta H_3$$

$$\Delta H_2$$

It is guaranteed to form a solution if the IMFs between solute - solvent is larger than the IMFs between solvent - solvent and solute - solute

Why do Solutions Form?

- A spontaneous process favors increase of entropy and decrease of enthalpy.
- When a solution forms: ΔS (+) and ΔH case dependant.



$$\Delta H_{\text{solution}} = \Delta H_{\text{solvent}} + \Delta H_{\text{solute}} + \Delta H_{\text{mix}} = \Delta H_{\text{hydration}} + \Delta H_{\text{solute}}$$

It is guaranteed to form a solution if the IMFs between mixture is larger than the IMFs between pure substance (solvent + solute)

Your Turn!

$$\Delta H_{\text{solution}} = \Delta H_{\text{hydration}} + \Delta H_{\text{solute}}$$

(+) (-) (+)

When NH_4NO_3 dissolves in water, the container gets cold. What can you conclude about the relative magnitudes of the absolute values of $\Delta H_{\text{hydration}}$ and ΔH_{solute} ?

A. $|\Delta H_{\text{hydration}}| > |\Delta H_{\text{solute}}|$

B. $|\Delta H_{\text{hydration}}| = |\Delta H_{\text{solute}}|$

C. $|\Delta H_{\text{hydration}}| < |\Delta H_{\text{solute}}|$

D. *Nothing can be concluded about the relative magnitude*

$$(-) = (-) + (+)$$

$$(-) > (+)$$



$$5 = (-1) + 6$$

$$5 = 3 + 2$$

Colligative Properties

- Physical properties of a solution depends only on the number (not the identity!) of solute particle.
- Vapor pressure lowering:

$$P_{\text{solution}} = \chi_{\text{solvent}} \times P_{\text{solvent}}^{\circ} \quad \left(\chi_{\text{solvent}} = \frac{n_{\text{solvent}}}{n_{\text{solvent}} + n_{\text{solute}}} \right)$$

$$\begin{aligned} \Delta P &= P_{\text{solvent}}^{\circ} - \chi_{\text{solvent}} \times P_{\text{solvent}}^{\circ} = P_{\text{solvent}}^{\circ} (1 - \chi_{\text{solvent}}) \\ &= P_{\text{solvent}}^{\circ} \times \left(\frac{n_{\text{solvent}} + n_{\text{solute}} - n_{\text{solvent}}}{n_{\text{solvent}} + n_{\text{solute}}} \right) = \underline{P_{\text{solvent}}^{\circ} \times \chi_{\text{solute}}} \end{aligned}$$

- Boiling point _____:

$$\Delta T_b = i \times K_b \times m_{\text{solute}} \quad (m = \underline{\hspace{2cm}})$$

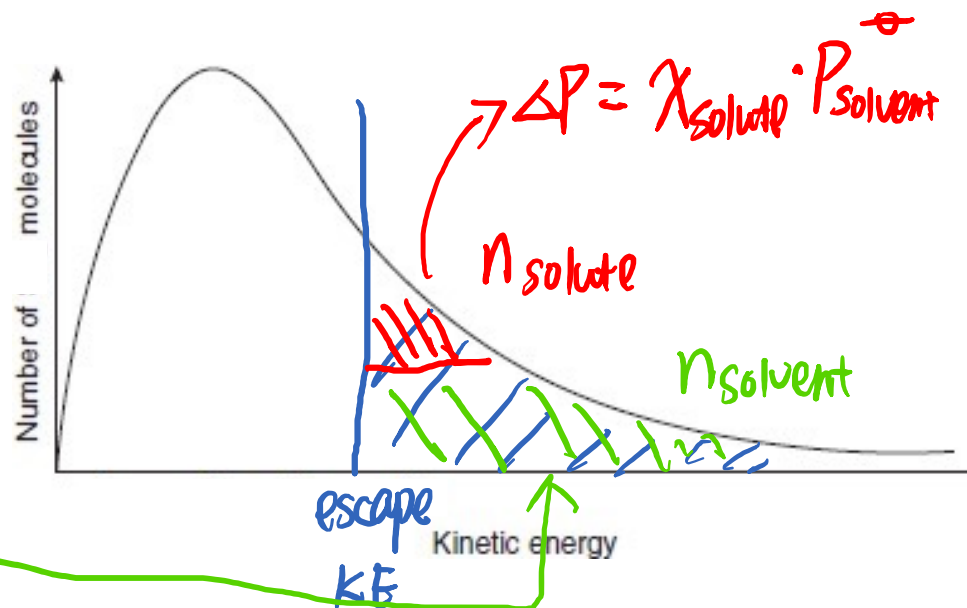
- Freezing point _____:

$$\Delta T_f = i \times K_f \times m_{\text{solute}} \quad (m = \underline{\hspace{2cm}})$$

Vapor pressure Lowering

- Electrolytes are not volatile.
- At a certain temperature, the number of particles that can escape to gas phase is the same between pure substance and solution.
- Most of solutes (solids) are not volatile and cannot escape to gas phase.
- # escaped molecule in solution < # escaped molecule in pure substance
- VP of solution < VP of pure substance

$$P_{\text{solution}} = \chi_{\text{solvent}} \times P_{\text{solvent}}^0$$

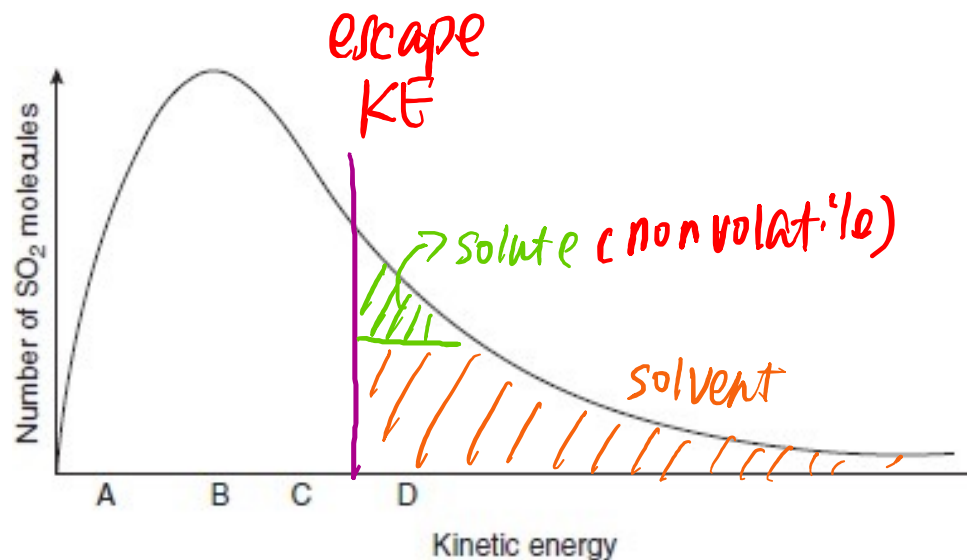


Vapor pressure Lowering

- Electrolytes are not volatile.
- At a certain temperature, the number of particles that can escape to gas phase is the same between pure substance and solution.
- Most of solutes (solids) are not volatile and cannot escape to gas phase.
- # escaped molecule in solution < # escaped molecule in pure substance
 $< 100\%$
 100%
- VP of solution < VP of pure substance

$$\Rightarrow P_{\text{solution}} = \chi_{\text{solvent}} \times P_{\text{solvent}}^0$$

$$\Delta P = \chi_{\text{solute}} P_{\text{solvent}}^0$$



Your Turn!

$$P_{\text{solution}} = \chi_{\text{solvent}} \times P_{\text{solvent}}^{\circ} \quad \Delta P = \chi_{\text{solute}} \times P_{\text{solvent}}^{\circ}$$

Calculate the decrease in vapor pressure ΔP after 218 g of glucose (molar mass = 180.2 g/mol) is dissolved in 460 mL of water (molar mass = 18.02 g/mol) at 30 °C. The vapor pressure of pure water at 30 °C is 31.82 mmHg. Assume the density of water is 1.00 g/mL.

1.4 mmHg

Your Turn!

$$P_{\text{solution}} = \chi_{\text{solvent}} \times P_{\text{solvent}}^0$$

Calculate the decrease in vapor pressure ΔP after 218 g of glucose (molar mass = 180.2 g/mol) is dissolved in 460 mL of water (molar mass = 18.02 g/mol) at 30 °C. The vapor pressure of pure water at 30 °C is 31.82 mmHg. Assume the density of water is 1.00 g/mL.

$$\Delta P = \chi_{\text{solute}} \times P_{\text{solvent}}^0$$

$$= \frac{218 \text{ g} \times \frac{1 \text{ mol}}{180.2 \text{ g}} \text{ glucose}}{218 \text{ g} \times \frac{1 \text{ mol}}{180.2 \text{ g}} + 460 \text{ mL} \times \frac{1.00 \text{ g}}{1 \text{ mL}} \times \frac{1 \text{ mol}}{18.02 \text{ g}}} \times 31.82 \text{ mmHg}$$

$$= 1.4 \text{ mmHg}$$

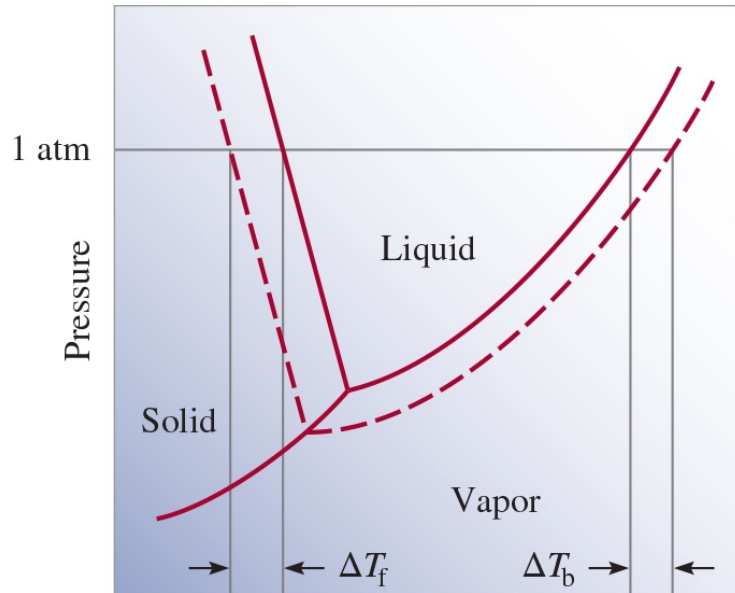
Freezing Point Depression & Boiling Point Elevation



For solution:
Solvents freeze as a pure solids.
Solutes keeps solvent in solution longer
Freezing Point of solution _____



For solution:
At the same T, vapor pressure ____
Need ____ T to vaporize all solvent
Boiling Point of solution _____



Compared to the pure solvent, the solution...
Triple point shift to _____

The normal boiling/freezing point of the solution:

$T_{\text{boiling}} =$

$T_{\text{freezing}} =$

van't Hoff Factor (i)

$$\Delta T_b = i \times K_b \times m_{\text{solute}}$$

$$\Delta T_f = i \times K_f \times m_{\text{solute}}$$

- i (van't Hoff Factor) = # particles solute per formula unit

For nonelectrolyte (eg. $\text{C}_6\text{H}_{12}\text{O}_6$): $i = \underline{\hspace{1cm}}$

For electrolyte (eg. Na_2SO_4): $i = \underline{\hspace{1cm}}$

Electrolytes have a ability to decrease f.p. or increase b.p.

- m (molality of the solute compound) =
- K_b and K_f are constant and depend on solvent. (Unit:)

Learning Check

Ethylene glycol (molar mass = 62.07 g/mol) is a common automobile antifreeze. It is water-soluble and fairly nonvolatile. Calculate the freezing boiling point of a solution containing 651 g of this substance in 2505 g of water. $K_{f,\text{water}} = 1.86^\circ\text{C}/m$

Your Turn! $\Delta T_f = i \times K_f \times m_{\text{solute}}$

Calculate the freezing point, in °C, of an antifreeze solution made up of 100.0 g ethylene glycol, $\text{C}_2\text{H}_6\text{O}_2$, (mm 62.07) and 100.0 g H_2O (mm=18.02). Ethylene glycol does not dissociate. $K_{f \text{ water}} = 1.86 \text{ }^\circ\text{C/m}$

Your Turn! $\Delta T_b = i \times K_b \times m_{\text{solute}}$

How many grams of NaCl (mm= 58.4 g/mol) must be dissolved in 255g H₂O (mm=18.02) to change the boiling point of water to 104.4 °C? $K_{b \text{ water}} = 0.51 \text{ } ^\circ\text{C/m}$.

If the #moles of sucrose = #moles of NaCl and the sucrose is added to the same volume of water, would the b.p. of the sucrose solution be the lower/same/higher as the b.p. of the NaCl solution?